

OPB Index Options Buying Bot

End-to-End System Review, Cleanup & Enhancement Report - v2.58.0 (2026-08-08)

1. Review Scope

Full audit of architecture, code, configuration, security, tests and documentation; live-market validation via a SHADOW-LIVE (SIGNAL_ONLY) session; generation of this summary and an architecture deck. No risk controls (SL/TARGET/drawdown), broker abstraction, or paper-mode invariants were modified.

2. Live-Market Session Status

Mode	SHADOW LIVE (SIGNAL_ONLY) - signals on real NSE market data, zero orders
Start time	2026-08-07 09:23 IST (market open 09:15 IST)
Process	index_trader.py, trading loop interval 30s, reconciliation CLEAN
Orders placed	0 (by design - no broker configured, gate enforced)
Live-readiness gate	BLOCKED - see section 5 (correct behaviour)

3. Verification Results

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4. Strengths

- Safety invariants:** Paper-mode invariant (simulated fills never reach a real broker), hard halt kill-switch, kill file, circuit breaker, watchdog thread, capital reservation lock, LTP sanity check.
- Governance depth:** Constitution Validation Engine (23 categories), pre-implementation compliance gate, AI governance gate, live-readiness scorecard (5 blocking criteria), release governance pipeline.
- Test maturity:** ~14,700 tests; dedicated suites for smoke, live-readiness, NSE recorder, constitution, config schema, paper trader, execution hardening.
- CI/CD completeness:** GitHub Actions with 9 jobs (lint, test matrix 3.11-3.14, coverage >=90%, security with pip-audit+bandit+semgrep, governance, certification, slow tests, build+checksum) + nightly full-suite cron.
- Deployment options:** Dockerfile + docker-compose + supervisord, Dockerfile.realestate, k8s configmap, launcher EXE with single-instance lock.
- Observability:** Prometheus metrics exporter, FastAPI enterprise dashboard (RBAC), health checker, audit trail (JSONL), structured logging with rotation + gzip.
- Security posture:** Secrets moved to OPBUYING_* env vars, SecureConfig redaction, config.local.json gitignored, Telegram auth allowlists + rate limits + audit, bandit in CI.

Multi-asset coverage: Equity, commodity (MCX), currency (CDS), index futures engines routed through a multi-asset dispatcher (equity enabled by default).

5. Weaknesses / Risks

Live-readiness gate unmet: 0/50 paper trades, 0 trading days, 0% win rate -> the gate correctly blocks LIVE (AUTO) start. No broker credentials configured (BROKER_API_ENABLED=false, GENERIC driver). A PAPER session runs daily to build the record.

Live data latency: NSE Akamai blocks automated scraping; yfinance fallback labels NIFTY LTP as 'last daily close' during live hours (stale intraday prices).

Dead-code register noise: 44,324 ORPHANED_SYMBOL entries, all MEDIUM; mostly test functions/constants (register regenerated 2026-08-08). No orphaned modules found, so no safe bulk pruning exists.

Static-analysis findings: Bandit: 0 HIGH / 0 MEDIUM / 298 LOW over 177,378 LOC (2026-08-08). All 14 MEDIUM resolved with proper fixes: identifier allow-lists on dynamic SQL, http/https scheme checks + opener-based URL calls, loopback default bind. Remaining 298 are LOW-severity heuristics (B101 asserts, B603 subprocess, B311 random) - see register.

Version skew: Local runtime Python 3.14.4 vs CI matrix 3.11/3.12. Startup gate allows 3.10-3.19, but CI should add 3.13/3.14.

Dashboard binding: web_dashboard binds 0.0.0.0 by default (opt-in feature; warns without TLS). Should default to 127.0.0.1.

ML cold start: ML classifier has no training data (0 trades); predict_win_prob degrades to 0.5 until the paper track record accumulates.

Config drift artifacts: Previous audit reports in docs/archive record config drift (EXECUTION_MODE defaults differ); current sync checks pass but older reports remain unarchived.

6. Live-Readiness Gate (blocks LIVE/AUTO start)

Criterion	Actual	Status
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Recommendation from the checker: continue paper trading until all blocking criteria are met. Score 1/5 blocking passed. A daily PAPER session is active to accumulate the 50-trade record.

7. Suggested Improvements (prioritised)

P1 - Build paper track record: Run EXECUTION_MODE=PAPER for >=10 trading days / >=50 trades to unblock the live-readiness gate. This is the single blocking item for AUTO.

P1 - Configure broker secrets: Set OPBUYING_BROKER_API_KEY/ACCESS_TOKEN/USER_ID/PASSWORD env vars and BROKER_DRIVER=KITE|ANGEL when ready; keep EXECUTION_MODE progression MANUAL->PAPER->SIGNAL_ONLY->AUTO.

P2 - DONE - Harden urlopen schemes: http/https scheme allowlists on all urlopen sites: core (6 modules) + scripts (realestate_synthetic_monitor, test_deployment); calls moved to build_opener().open() - no nosec.

P2 - DONE - Fix dynamic SQL: Identifier allow-list validation (_safe_ident) on all dynamic table/column SQL in scripts (check_db_integrity, migrate_to_postgresql, verify_restore, quantitative_validation_report); values parameter-bound - no nosec. Bandit MEDIUM 14 -> 0.

P2 - DONE - Dashboard default bind: web_dashboard + enterprise_dashboard + launch_realestate bind 127.0.0.1 by default (0.0.0.0 only when explicitly configured).

P2 - DONE - CI matrix: Python 3.13/3.14 added to the GitHub Actions test matrix; semgrep added to the security job; nightly full-suite cron scheduled.

P3 - Register hygiene: Run scan_dead_code.py --update-registers after every scan so the register reflects current code (stale rows were regenerated 2026-08-08).

P3 - Archive stale docs: Move 2026-07-20 audit reports in docs/archive to a dated subfolder and regenerate config-drift reports against current defaults.